

STAT 626 Homework 1

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```
library(astsa)
```

I.

1.1)

Generating 100 observations from $x_t = .9x_{t-2} + w_t$; $\sigma_w = 1$:

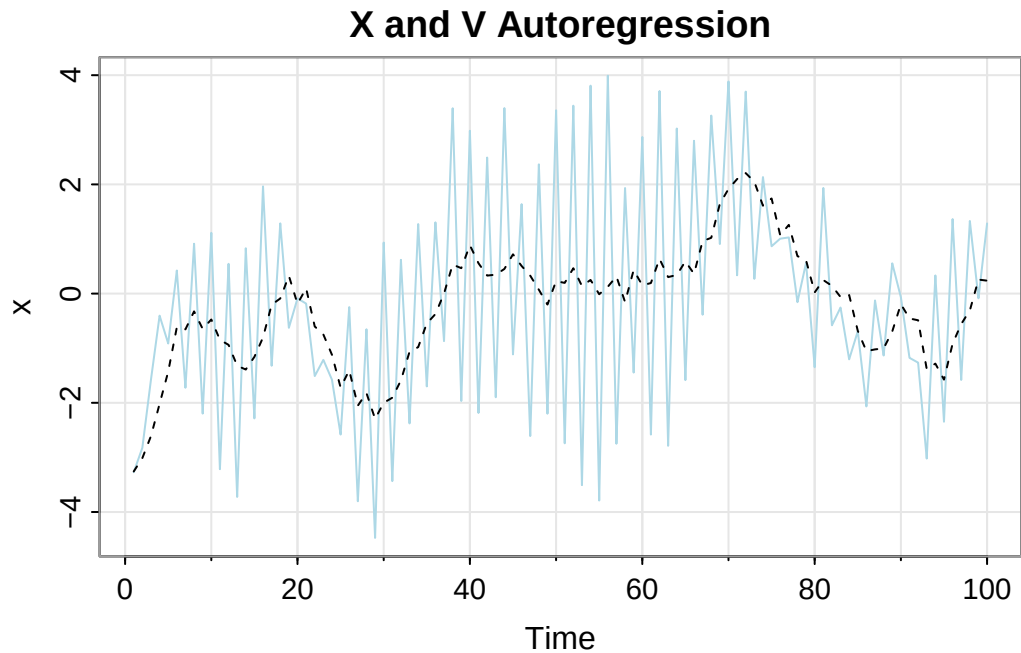
```
set.seed(10)
w = rnorm(100 + 50)
#Extra 50 for Startup
x_long = filter(w, filter = c(0,.9), method = "recursive")
```

Applying moving average filter $v_t = (x_t + x_{t-1} + x_{t-2} + x_{t-3})/4$:

```
#Extra 50 for Startup
v = filter(x_long, sides = 1, filter = rep(1/4,4))[-(1:50)]
x = x_long[-(1:50)]
```

Plotting:

```
tsplot(x, col = "lightblue", main = "X and V Autoregression")
lines(v, lty = "dashed")
```



b)

Generating $x_t = 2 \cos(2\pi t/4) + w_t$:

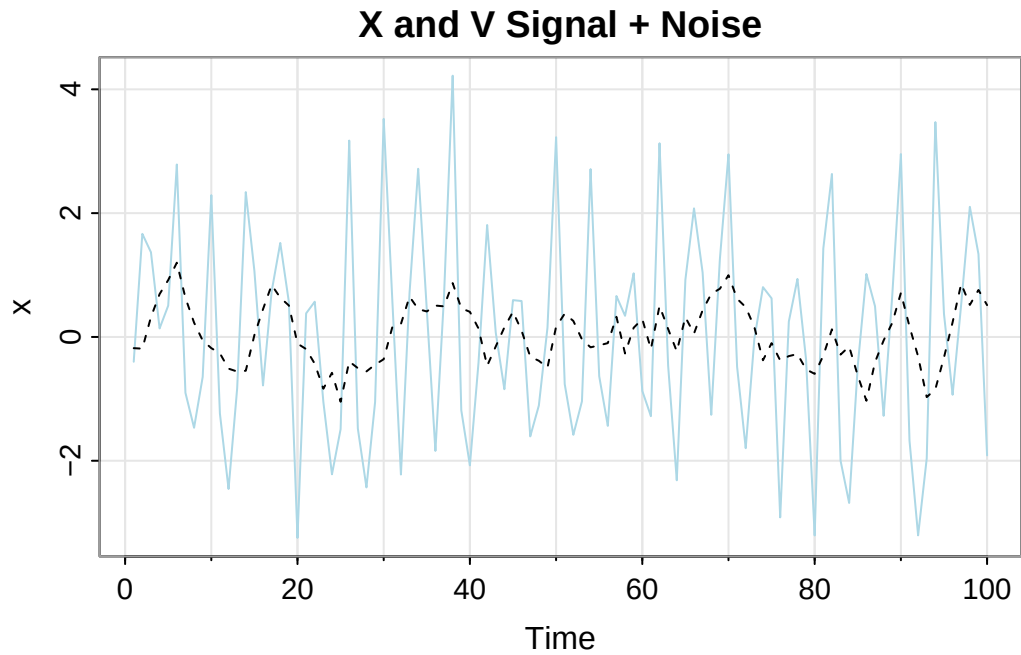
```
#Extra 50 for startup
t = 1:150
cs = 2*cos(2*pi*t/4)
x_long = cs + w
```

Applying moving average filter $v_t = (x_t + x_{t-1} + x_{t-2} + x_{t-3})/4$:

```
#Extra 50 for Startup
v = filter(x_long, sides = 1, filter = rep(1/4,4))[-(1:50)]
x = x_long[-(1:50)]
```

Plotting:

```
tsplot(x, col = "lightblue", main = "X and V Signal + Noise")
lines(v, lty = "dashed")
```



c)

Loading log of J&J data:

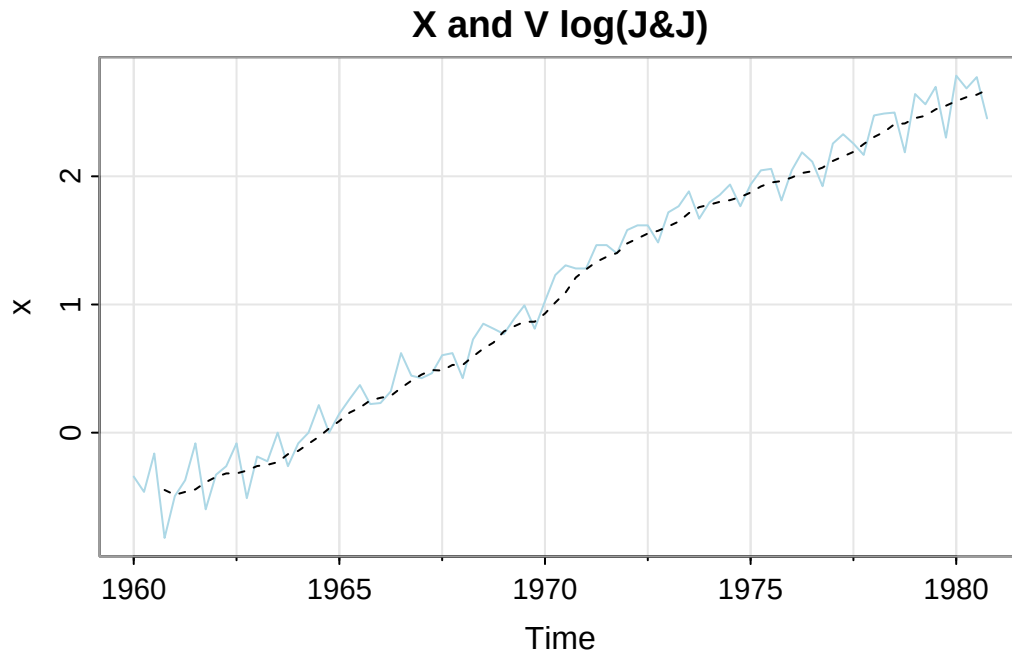
```
x = log(jj)
```

Applying moving average filter $v_t = (x_t + x_{t-1} + x_{t-2} + x_{t-3})/4$:

```
#Extra 50 for Startup
v = filter(x, sides = 1, filter = rep(1/4,4))
```

Plotting:

```
tsplot(x, col = "lightblue", main = "X and V log(J&J)")
lines(v, lty = "dashed")
```



d)

According to Investopedia: “Seasonal adjustments are a statistical method to smooth out aberrations in time series of certain types of economic activity that occur on a regular or cyclical basis. These adjustments provide a clearer view of net trends and nonseasonal changes in data.”

In the case of J&J, we have previously analyzed the fact that there is a pattern of three quarters of similar performance and a quarter of poorer performance. By using a seasonal adjustment we can focus on the overall trend of the company, and not the seasonal effect causing that deviation.

e)

I learned that using a moving average filter can help us visually identify trends in data. This is because variance is reduced by averaging.

With this smaller variance, high frequency behavior is removed. A good example of this is in the Signal + Noise data, we can now see some evidence of the Cosine curve which is generating the data set while before the quick oscillations up and down made it harder to identify.

In the J&J case, we can use the averaging of 4 observations to intentionally remove the expected seasonality and only analyze the overall yearly trend in earnings.

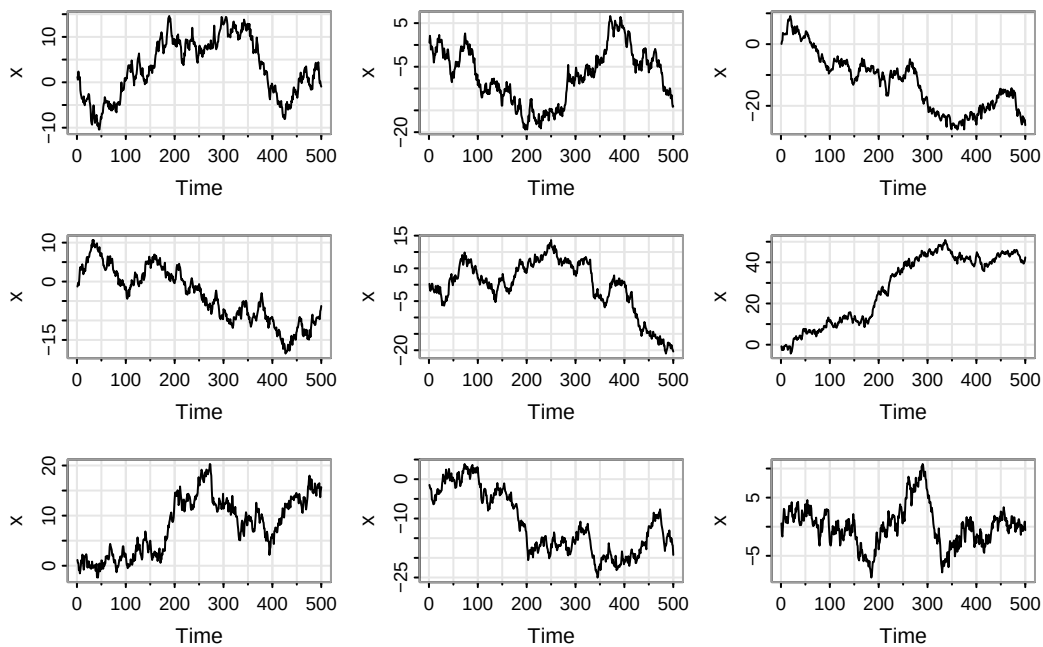
In the auto regression case we can more easily see the periodical ups and downs that exist due to each observation being based on previous observations.

1.3)

a)

Random walk plots:

```
par(mfrow = c(3,3))
set.seed(20)
for(i in 1:9){
  w = rnorm(500)
  x = cumsum(w)
  tsplot(x)
}
```



b)

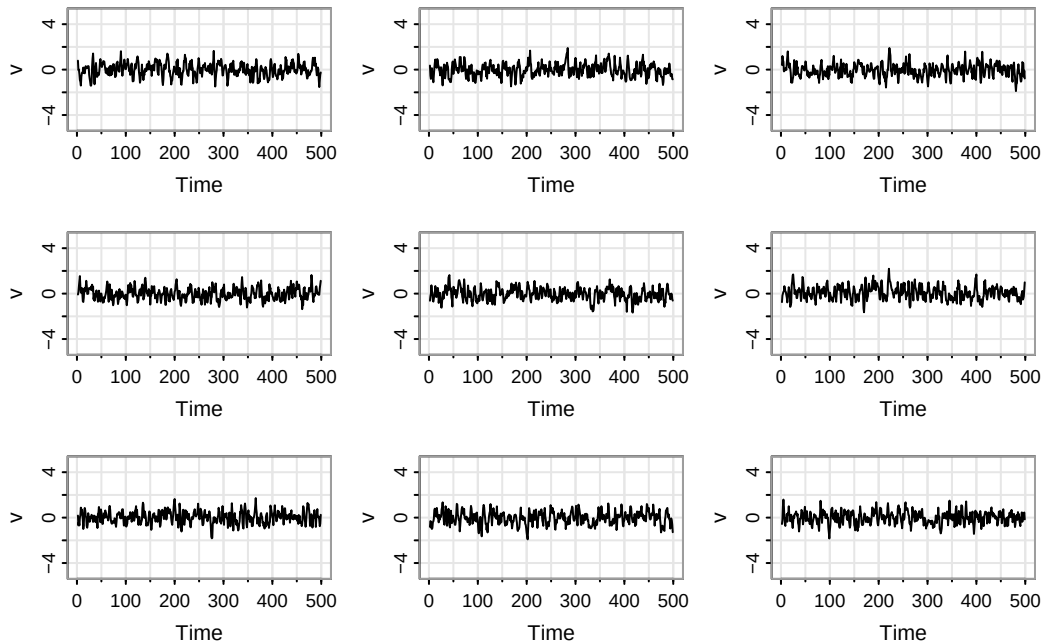
Moving Average:

```
par(mfrow = c(3,3))
set.seed(20)
for(i in 1:9){
  w = rnorm(500)
  v = filter(w, sides = 2, filter = rep(1/3,3))
}
```

```

    tsplot(v, ylim = c(-5,5))
}

```



c)

For random walks we see that each value is dependent on the previous value, and the path each random walk takes can differ quite a bit. It is also tempting to read a pattern into each plot, but we know that each new value is simply the previous value + a random factor.

For moving averages, all plots look similar. The plots are generally centered around 0, with oscillations of smaller magnitude due our averaging.

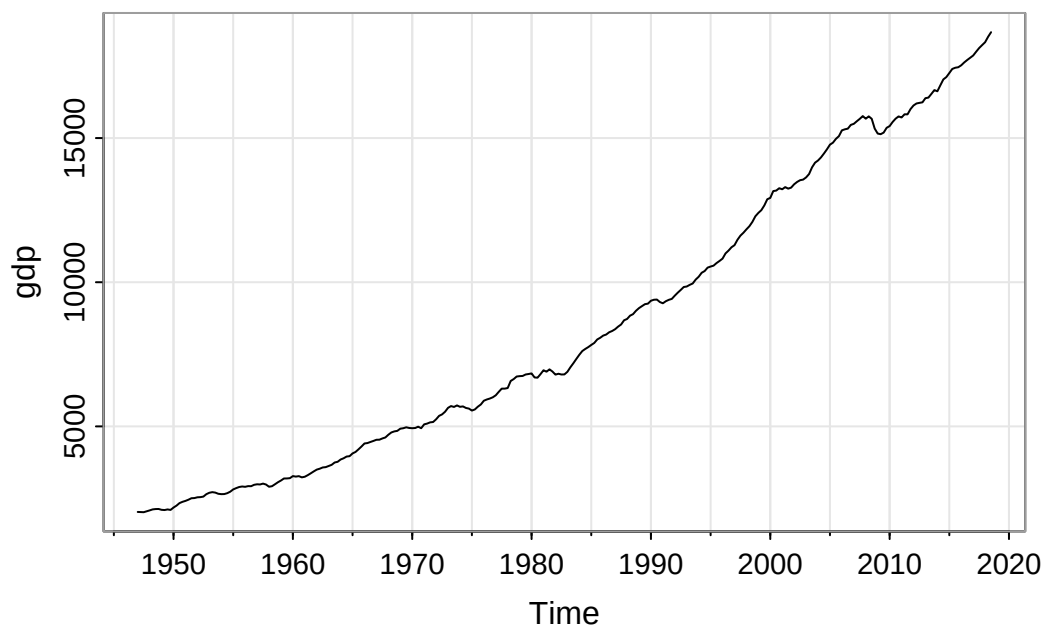
1.4)

a)

```

    tsplot(gdp)

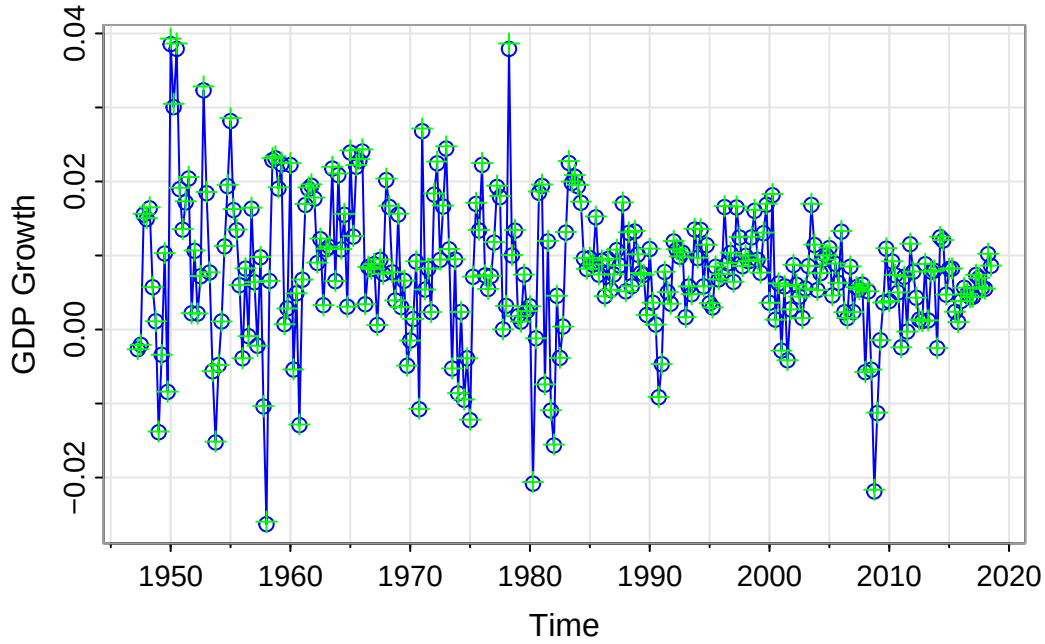
```



This looks like a random walk with a positive drift component to me.

b)

```
tsplot(diff(log(gdp)), type = "o", col = "blue", ylab = "GDP Growth")  
points(diff(gdp)/lag(gdp, -1), pch = 3, col = "green")
```



The differences between the two methods are very small, they only would get large if the rate of change was very large over a time period. We don't see that here.

c)

Between random walks and moving averages, the moving average model is clearly more appropriate. The growth rate is centered around about 0.01, has oscillations which are not large in nature and nearby points are close in magnitude.

1)

Showing $\sum_{i=1}^n (x_i - \bar{x}) = 0$:

$$\text{using } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x} = n\bar{x} - n\bar{x} = 0$$

Showing $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n (x_i - \bar{x})y_i$:

$$\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n (x_i - \bar{x})y_i - \bar{y} \sum_{i=1}^n (x_i - \bar{x})$$

The right term is equal to 0, as we have previously shown that $\sum_{i=1}^n (x_i - \bar{x}) = 0$.

2)

Showing $S_x^{-2} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n c_i y_i$ where $c_i = \frac{x_i - \bar{x}}{S_x^2}$, $S_x^2 = \sum_{i=1}^n (x_i - \bar{x})^2$:

Our previous result of $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n (x_i - \bar{x})y_i$ thus leaving us with:

$$S_x^{-2} \sum_{i=1}^n (x_i - \bar{x})y_i = \sum_{i=1}^n c_i y_i$$

3)

a)

We have the residual sum of squares equation:

$$Q(\beta_0, \beta_1) = \sum (y_i - \beta_0 - \beta_1 x_i)^2$$

We are looking for the (β_0, β_1) that minimize this equation, so we take the derivative with respect to each:

$$\frac{\partial Q(\beta_0, \beta_1)}{\partial \beta_0} = -2 \sum (y_i - \beta_0 - \beta_1 x_i)$$

$$\frac{\partial Q(\beta_0, \beta_1)}{\partial \beta_1} = -2 \sum x_i (y_i - \beta_0 - \beta_1 x_i)$$

We set each equal to zero and solve the system of equations. Starting with the partial derivative with respect to β_0 :

$$-2 \sum (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$-\sum y_i + \sum \beta_0 - \beta_1 \sum x_i = 0$$

$$\beta_0 = \frac{\sum y_i + \beta_1 \sum x_i}{n} = \bar{y} - \beta_1 \bar{x}$$

Taking β_0 and substituting into the partial derivative with respect to β_1 :

$$-2 \sum x_i(y_i - \bar{y} + \beta_1 \bar{x} - \beta_1 x_i) = 0$$

$$\sum x_i(y_i - \bar{y}) + \beta_1 \sum x_i(\bar{x} - x_i) = 0$$

$$\beta_1 = \sum \frac{x_i(y_i - \bar{y})}{x_i(x_i - \bar{x})}$$

To get this into the usual form we need to do some algebra. Starting with the denominator:

$$\sum x_i(x_i - \bar{x}) = \sum x_i^2 - \sum x_i \bar{x} = \sum x_i^2 - n\bar{x}^2 = \sum (x_i^2 - \bar{x}^2)$$

This term is equivalent to $\sum (x_i - \bar{x})^2 = \text{SXX}$, shown below:

$$\sum (x_i - \bar{x})^2 = \sum (x_i^2 - 2x_i \bar{x} + \bar{x}^2) = \sum (x_i^2 - 2\bar{x}^2 + \bar{x}^2) = \sum (x_i^2 - \bar{x}^2)$$

The numerator $\sum x_i(y_i - \bar{y}) = \sum (x_i - \bar{x})(y_i - \bar{y})$ using results from part 1.

So our LSE estimates are:

$$\hat{\beta}_0 = \bar{y} - \beta_1 \bar{x}$$

$$\hat{\beta}_1 = \sum \frac{(x_i - \bar{x})(y_i - \bar{y})}{(x_i - \bar{x})^2} = \frac{\text{SXY}}{\text{SXX}}$$

b)

We can rewrite $\hat{\beta}_1$ as $S_x^{-2} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ which is identical to part 2

With assuming $\epsilon_i \sim N(0, \sigma^2)$ we can determine the distribution $\hat{\beta}_1$. First I substitute in the assumption of $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$

$$\hat{\beta}_1 = \sum_{i=1}^n c_i (\beta_0 + \beta_1 x_i + \epsilon_i)$$

We assume that x_i are chosen, and thus fixed variables without a distribution. So, since these are constants we can end up with a shifted normal distribution. To find μ we take the expected value of $\hat{\beta}_1$:

$$E[\hat{\beta}_1] = \sum_{i=1}^n c_i(\beta_0 + \beta_1 x_i)$$

Breaking apart c_i helps further simplify:

$$\sum \frac{(x_i - \bar{x})(\beta_0 + \beta_1 x_i)}{(x_i - \bar{x})^2} = \sum \frac{(x_i - \bar{x})\beta_0}{(x_i - \bar{x})^2} + \sum \frac{\beta_1(x_i - \bar{x})x_i}{(x_i - \bar{x})^2}$$

Now we use our results from part 1. Starting with the left term, we have shown that $\sum(x_i - \bar{x}) = 0$, so it is equal to zero. The right term utilizes our second part 1 result. So we can say:

$$= \sum \frac{\beta_1(x_i - \bar{x})^2}{(x_i - \bar{x})^2} = \beta_1$$

Now finding the variance of the normal distribution of $\hat{\beta}_1$ is rather simple:

$$Var(\hat{\beta}_1) = Var\left(\sum_{i=1}^n c_i(\beta_0 + \beta_1 x_i + \epsilon_i)\right) = Var\left(\sum_{i=1}^n c_i \epsilon_i\right) = \sum_{i=1}^n c_i^2 \sigma^2 = \frac{\sum (x_i - \bar{x})^2}{(\sum (x_i - \bar{x})^2)^2} \sigma^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sigma^2}{SSX}$$

So, the distribution of $\hat{\beta}_1$ is $N(\beta_1, \sigma^2/SSX)$.

c)

$$e_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

From previous work we have $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ so:

$$e_i = y_i - \bar{y} + \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i$$

So:

$$\sum_{i=1}^n e_i = \sum_{i=1}^n (y_i - \bar{y}) - \hat{\beta}_1 \sum_{i=1}^n (x_i - \bar{x})$$

The result $\sum_{i=1}^n e_i = 0$ then follows directly from our first result in part 1.

Building off this we can show the next result:

$$\sum_{i=1}^n e_i x_i = \sum_{i=1}^n x_i (y_i - \bar{y}) - \hat{\beta}_1 \sum_{i=1}^n x_i (x_i - \bar{x})$$

We use the second result from part 1 twice:

$$= \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) - \hat{\beta}_1 \sum_{i=1}^n (x_i - \bar{x})^2 = \text{SXY} - \hat{\beta}_1 \text{SXX}$$

We then substitute in the estimate for $\hat{\beta}_1$:

$$= \text{SXY} - \frac{\text{SXY}}{\text{SXX}} \text{SXX} = 0$$

The geometrical interpretation is that the vector of residuals e is orthogonal to column space of the design matrix x .